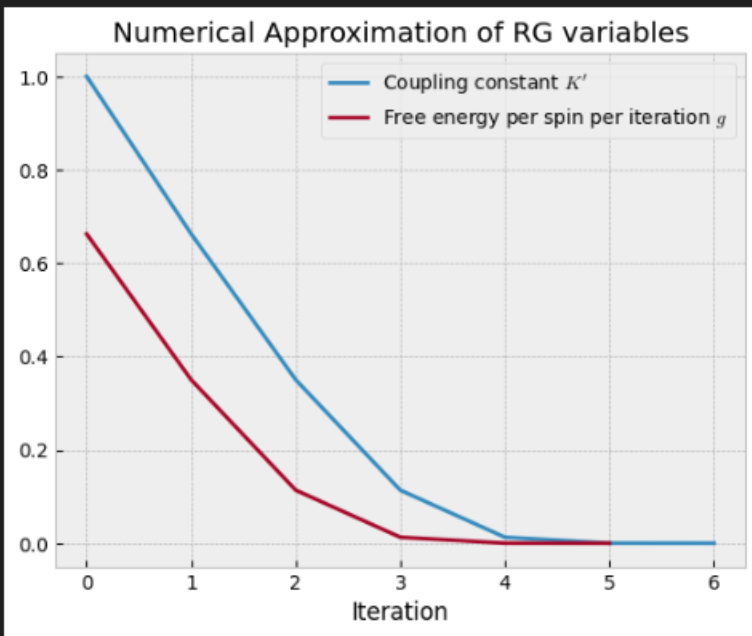




```
# As we can easily see the limits of both series is towards zero K. That is due to the fact
# that the point K is a stable attractive point for the system
```

[5] ✓ 0.0s

```
# b) From the lectures we know
```

```
def freeEnergyRG(array):
    sumVar = 0
    for index,g in enumerate(array):
        sumVar += 2** (-index) * g
    return sumVar
```

```
def freeEnergyTrue(K):
    return np.log(2*np.cosh(K))
```

[43] ✓ 0.0s

```
print(f"For the RG approximation we get {freeEnergyRG(gArray):0.3f}, whilst the true result is : {freeEnergyTrue(KappaArray[0]):0.3f}")
```

[48] ✓ 0.0s

```
# There is a bit of mismatch between the results, but it is to be expected as
# 1) We have not performed many iterations
# 2) We are far away from a stable point, which would guarantee faster convergence
```

```
#Exercise 2 Computer simulation of the Ising model
```

```
# a)
```

```
# Helpfull functions
```

```
def generateStartingPopulation(Nx,Ny):
```

```
    return np.random.choice([1, -1], size=(Nx, Ny)) #Thank God numpy has that command
```

```
def pickRandomPosition(Nx,Ny):
```

```
    return (np.random.randint(0, Nx), np.random.randint(0, Ny))
```

```
def computeEnergyChange(i,j,Nx,Ny,population):
```

```
    # This is a bit tricky to implement due to the boundary conditions, but we march forward
```

```
    # The way we are gonna do this is generate an array of all the positions the algo should check
```

```
    # This approach allows for an elegant implementation of the periodic bounds, with the modulo operator
```

```
    checkArray = [ ( (i-1) % (Nx), j) , #Left of center
                   ( (i+1) % (Nx), j) , #Right of center
                   ( i, (j-1) % (Ny) ) , #Down of center
                   ( i, (j+1) % (Ny) ) , #Up of center
                   ]
```

```
    #The reason we take the size of each lattice dimension, is because arrays start counting at 0, not 1
```

```
    changeInEnergy = 0
```

```
    for neighbour in checkArray:
```

```
        changeInEnergy += + population[i][j]* population[neighbour[0]][neighbour[1]] #If we did the spin flip
```

```
    return 2*changeInEnergy #  $DE = E_f - E_i = 2 * E_f$ 
```

```
def runSimulation(startPopulation,temperature, stepsMC):
    Nx, Ny = len(startPopulation[0]), len(startPopulation)
    N, MagnetizationArray, populations = Nx * Ny, [], []
    population = startPopulation
    for index in range(stepsMC):
        for step in range(N):
            i,j = pickRandomPosition(Nx,Ny)
            energyChange = computeEnergyChange(i,j, Nx, Ny,population)
            if energyChange <= 0 or (np.random.uniform(0,1) <= np.exp(-energyChange / temperature) ):
                population[i][j] = population[i][j] * (-1)
            MagnetizationArray.append(np.sum(population)/N)
            populations.append(np.copy(population))
    return np.array(MagnetizationArray),populations
```

✓ 0.0s

The Monte Carlo simulation favors motion to an equilibrium, by immediately flipping the spins in order to drive the system into a lower energy state. Also, it allows for small deviations from such a state, by employing random flipping, so as to ensure that the system explores more areas of the phase space, that may lead to more optimal solutions. In that regard, by having a computationally cheap function that penalizes the more energy it would take for them to flip, it guarantees a balance between exploration and exploitation of the phase space optima.

✓ 0.0s

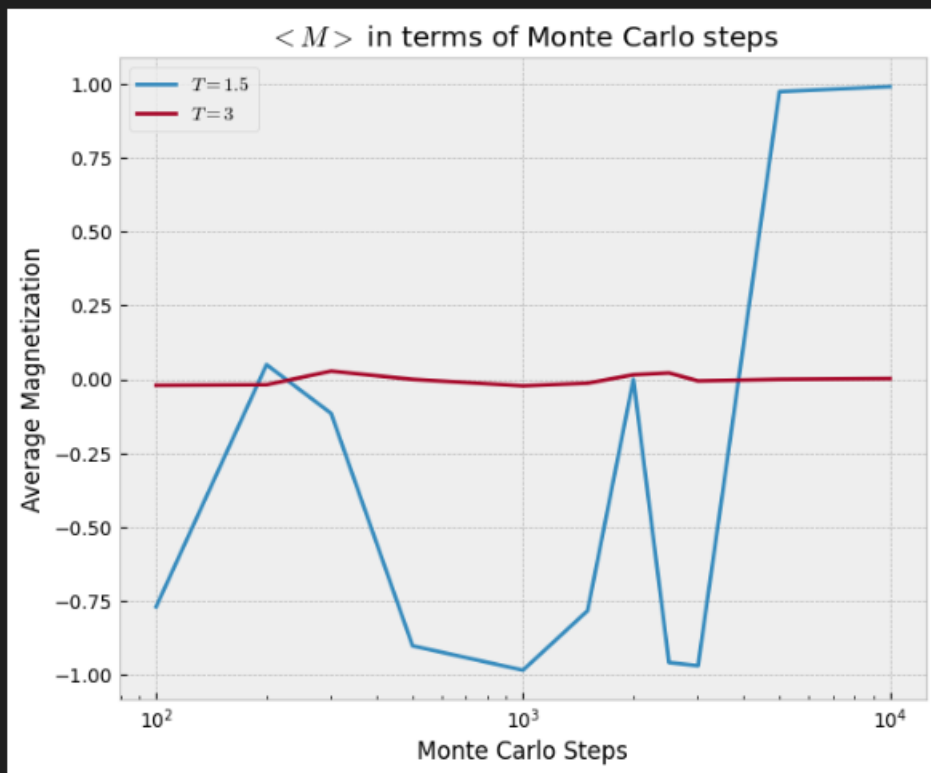
b)
The Monte Carlo simulation is a probabilistic numerical approach to evaluating efficiently large phase spaces. As per the law of large numbers and the central limit theorem, the random guess the simulation returns must at some point resemble a Gaussian distribution with mean μ and variance σ . Therefore, we get a result with relative error, $\sigma/(\mu * \sqrt{N})$. That means we expect the results we get to scale better with accuracy $O(\sqrt{N})$. As we need large numbers relative to the dimensions of the grid, we expect to get that above $O(1000)$.

✓ 0.0s

```
stepsIntervals = [100,200,300,500,1000,1500,2000,2500,3000,5000,10000]
lowTempResults = [np.average(runSimulation(generateStartingPopulation(32,32),1,step)[0]) for step in stepsIntervals]
highTempResults = [np.average(runSimulation(generateStartingPopulation(32,32),3,step)[0]) for step in stepsIntervals]
```

✓ 6m 34.1s

```
plt.figure(figsize=(7.5, 6), dpi=100)
plt.plot(stepsIntervals, lowTempResults, label=r'$T = 1.5$ ')
plt.plot(stepsIntervals, highTempResults, label=r'$T = 3$ ')
plt.legend()
plt.xlabel('Monte Carlo Steps')
plt.ylabel('Average Magnetization')
plt.title(r'$\langle M \rangle$ in terms of Monte Carlo steps')
plt.xscale('log')
plt.show()
```



```
# As it can be seen, the Monte Carlo approach converged rather nicely for the high-temperature case,
# but shows chaotic behaviors inherent in the low-temperature system. Nevertheless,
# after enough steps, it manages to converge to the proper result.
```

```

def createGIF(simulation, temperature, M):
    cmap = plt.cm.coolwarm
    bounds = [-1, 0, 1]
    norm = mcolors.BoundaryNorm(bounds, cmap.N)

    # List to store generated frames
    frames = []

    for i, (data,m) in enumerate(zip(simulation,M)):
        # Create a figure
        fig, ax = plt.subplots()
        im = ax.imshow(data, cmap=cmap, norm=norm)
        ax.set_title(rf"${M} = {m:0.5f} $ , $T = {temperature:0.2f}$")
        ax.set_xlabel("X-axis")
        ax.set_ylabel("Y-axis")
        ax.invert_yaxis()
        plt.colorbar(im, ax=ax, label="Spin")

        # Save frame to a temporary file
        temp_filename = f"frame_{i}.png"
        plt.savefig(temp_filename)
        frames.append(temp_filename)
        plt.close(fig)

    # Create GIF from frames
    gif_filename = f"GIF_for_T_{temperature:0.2f}.gif"
    with imageio.get_writer(gif_filename, mode="I", duration=0.5) as writer:
        for frame in frames:
            image = imageio.v3.imread(frame)
            writer.append_data(image)
            os.remove(frame)
    print("GIF saved as output.gif")
    pass

```

12] ✓ 0.0s

```

#Running the simulation for all the different temperatures of interest
# c)
Tarray = np.linspace(0.5,3,30)
TempsToGIF = [T for T in Tarray]
magnetizationTotalArray, gifSave = [], []
for T in tqdm(Tarray):
    magnetizationArray, populations = runSimulation(generateStartingPopulation(32,32),T,3000)
    if T in TempsToGIF:
        gifSave.append(populations)
    magnetizationTotalArray.append(magnetizationArray)

```

13] ✓ 11m 41.3s

100% 30/30 [11:41<00:00, 23.38s/it]

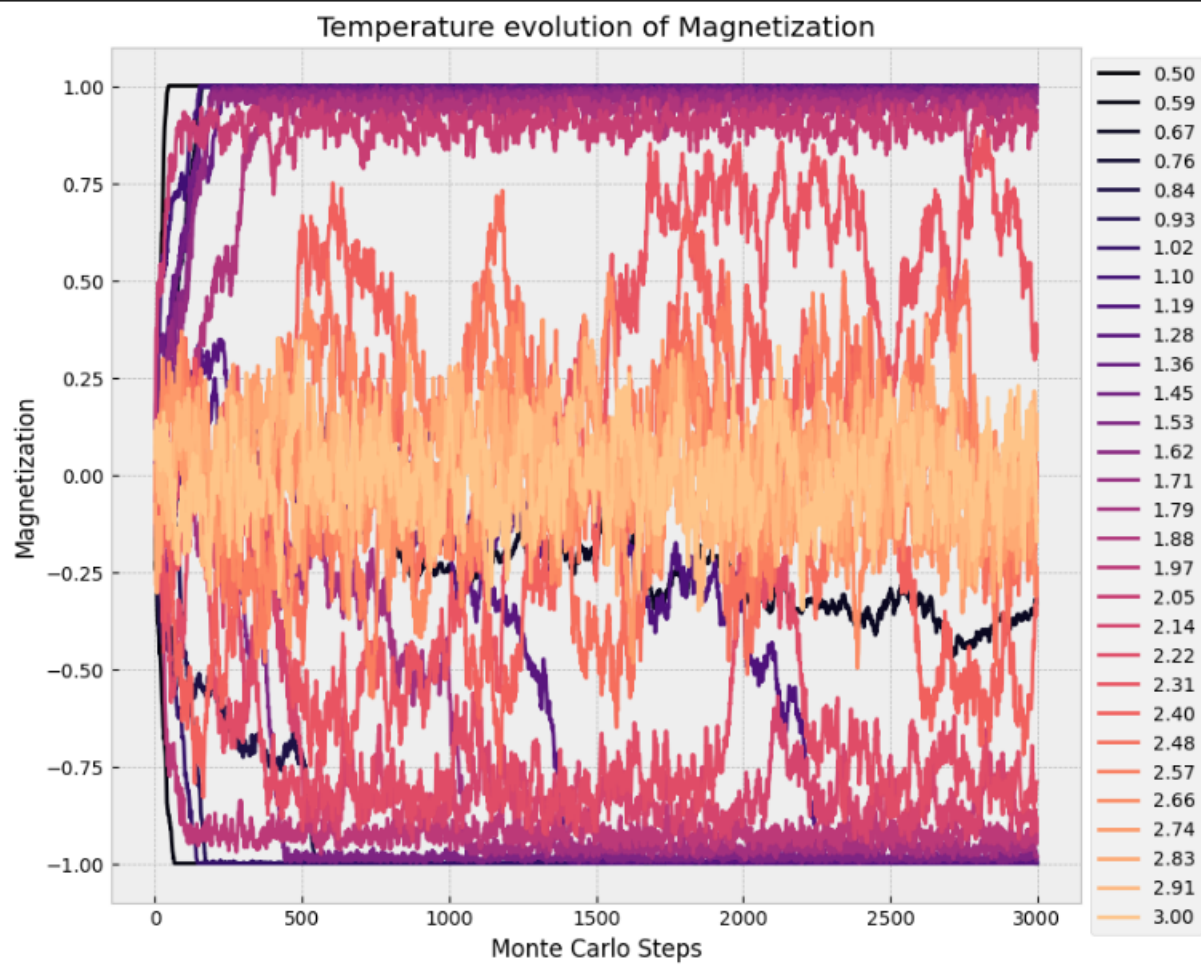
```

cmap = plt.get_cmap('magma')

# Normalize the index to be between 0 and 1 for the colormap
norm = plt.Normalize(vmin=0, vmax=len(magnetizationTotalArray)+3)
plt.figure(figsize=(9, 8), dpi=100)
for index, (magnetization, temp) in enumerate(zip(magnetizationTotalArray, Tarray)):
    color = cmap(norm(index))
    plt.plot(magnetization, label=f'{temp:.2f}', color=color)
plt.legend(loc='upper left', bbox_to_anchor=(1, 1))
plt.xlabel('Monte Carlo Steps')
plt.ylabel('Magnetization')
plt.title('Temperature evolution of Magnetization')
plt.show()

```

[15] ✓ 0.3s



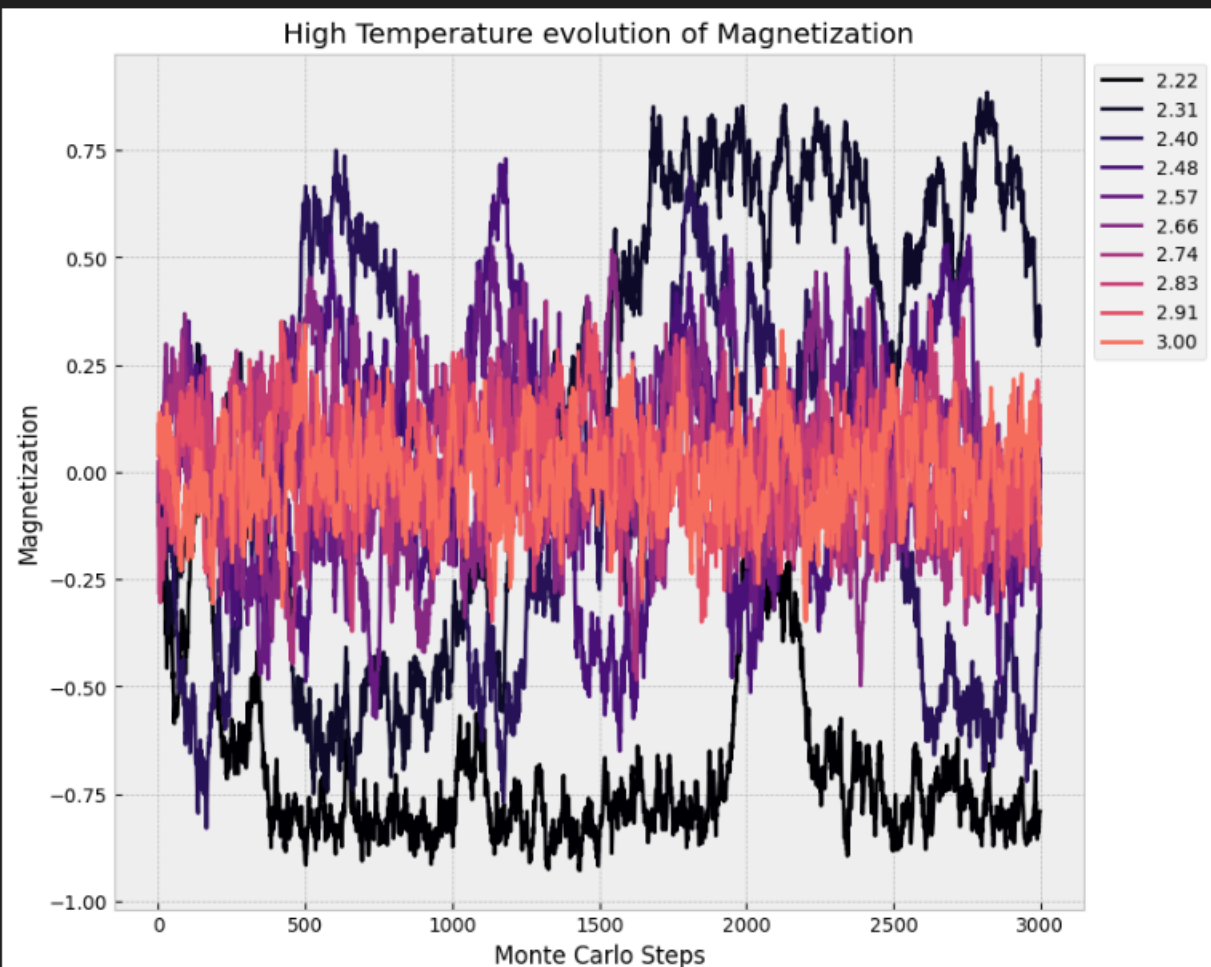
```

cmap = plt.get_cmap('magma')

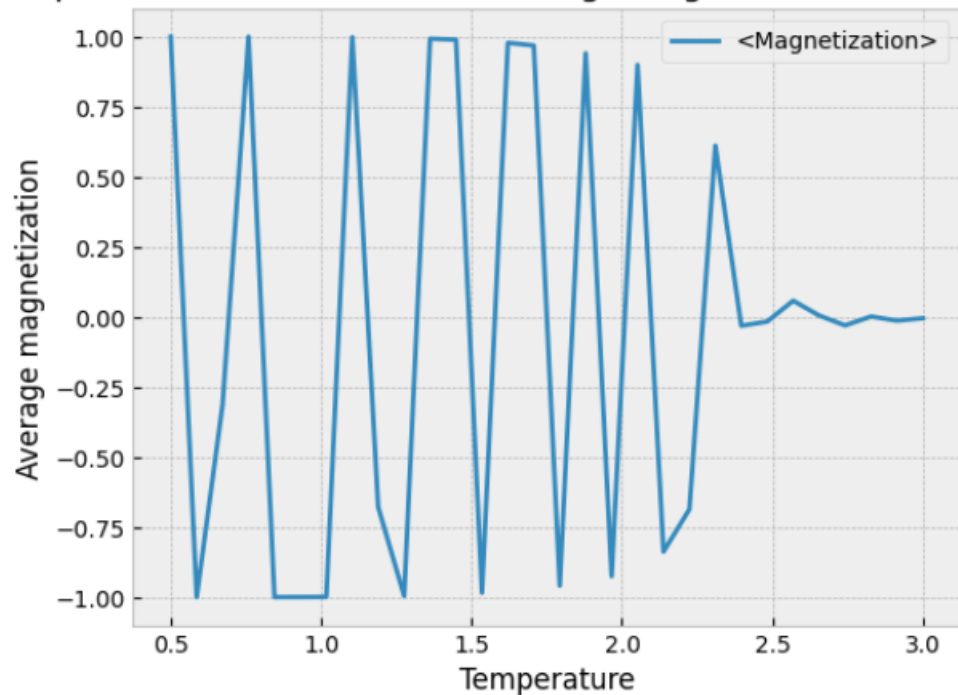
# Normalize the index to be between 0 and 1 for the colormap
norm = plt.Normalize(vmin=0, vmax=len(magnetizationTotalArray[20::])+3)
plt.figure(figsize=(9, 8), dpi=100)
for index, (magnetization, temp) in enumerate(zip(magnetizationTotalArray[20::], Tarray[20::])):
    color = cmap(norm(index))
    plt.plot(magnetization, label=f'{temp:.2f}', color=color)
plt.legend(loc='upper left', bbox_to_anchor=(1, 1))
plt.xlabel('Monte Carlo Steps')
plt.ylabel('Magnetization')
plt.title('High Temperature evolution of Magnetization')
plt.show()

```

[18] ✓ 0.1s



Temperature evolution of the average Magnetization in the system

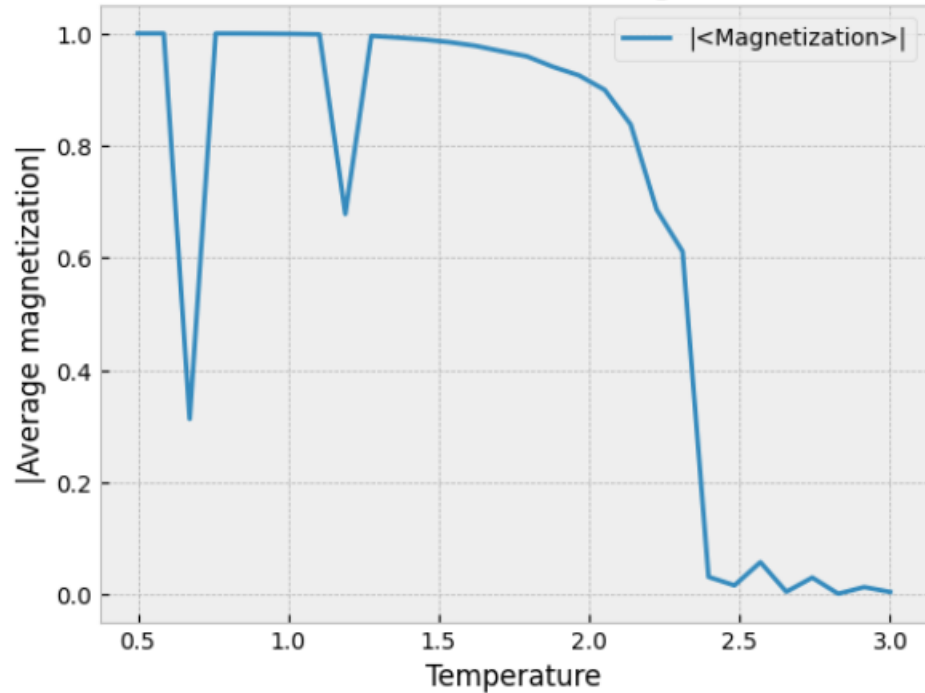


```
# For the high-temperature regime, the spin orientation is dominated by random fluctuations
# that result in the system having no preference in the number of up or down spins.
# Thus, the total magnetization is zero.
# For the low-temperature regime, the magnetic field of the different areas and particles dominates
# the system interactions. The system urgently wants to find a stable energy. In doing so, any
# starting island or majority of spins dominates the dynamics, and the system ends up either at 1, -1, or 0.
# Our graph truly encapsulates such behavior, and we can confidently confirm that the Monte Carlo
# simulation is an accurate representation of the Ising model in 2D.
```

```
plt.plot(Tarray,np.abs(averages),label='|<Magnetization>|')
plt.xlabel('Temperature')
plt.ylabel('|Average magnetization|')
plt.title('Temperature evolution of the absolute <Magnetization> in the system')
plt.legend()
plt.show()
```

[55] ✓ 0.0s

Temperature evolution of the absolute <Magnetization> in the system



Clearly our system has a phase transition at temperature of around 2.25 as was expected from the 2D Ising model

You can find the full code and some interesting GIFs at <https://github.com/grk38/Theoretical-Statistical-Mechanics-11/tree/main>

[59] ✓ 0.0s